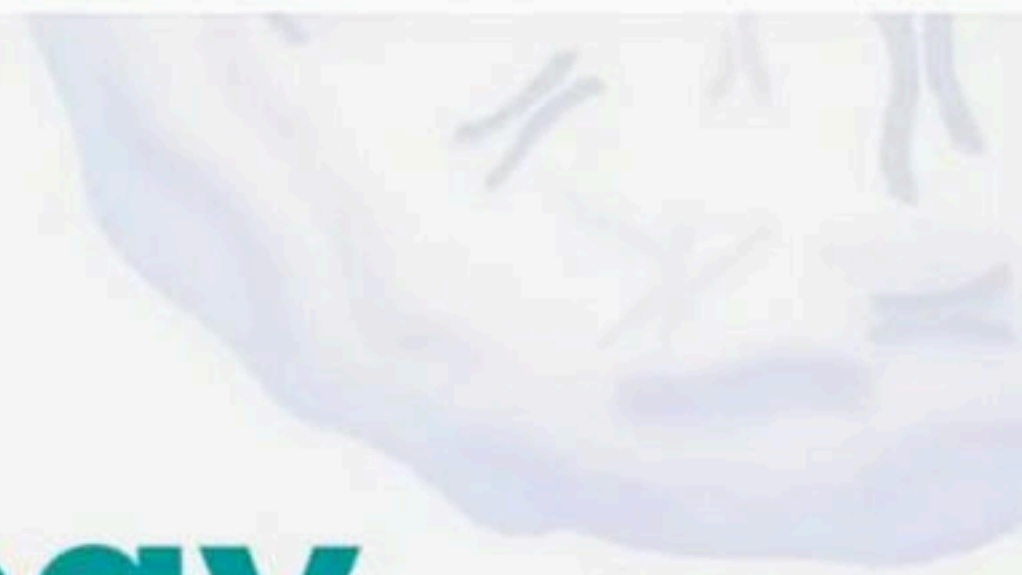


Biotechnology : Application - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



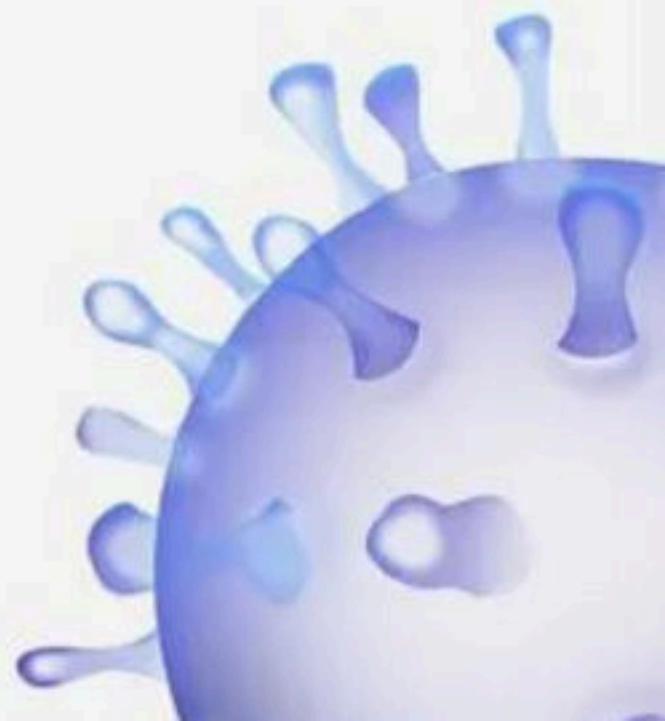
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Biotechnology Application ✓



BIOLOGYLIVE



1. crylab and cryIIAb produce toxins that control

2025

- (1) Cotton bollworms and corn borer, respectively.
- (2) Corn borer only
- (3) Cotton bollworms only
- ✓ (4) Corn borer and Cotton bollworms, respectively



2. Which body of government of India regulates GM research and safety of introducing GM organisms for public services?

- (1) Research committee on Genetic manipulation
- (2) Bio-safety committee
- ✓ (3) Genetic engineering approval committee ✓
- (4) Indian council of Agricultural research



3. Plants in comparison to animals are more rapidly manipulated by genetic engineering, which is the most appropriate reason for this character.

- (1) Totipotency is shown by plant cells ✓
- (2) Single somatic cell can regenerate whole plant body. ✓
- (3) Genetic engineering is supplemented with tissue culture technique ✓
- ✓ (4) All the above



4. The first transgenic cow, Rosie produced

- ✓ (1) Human protein enriched milk (2.4 g/L)
- (2) Human protein enriched milk (4.2 g/L)
- (3) Human protein enriched milk (2.4 g/mL) ✗
- (4) Human protein enriched milk (4.2 g/mL)



5. The genes obtained from B. thuringiensis is called

- ✓ (1) Cry gene
- (2) Nif gene
- (3) Bac gene
- (4) Bt gene ✗



6. Some of the steps involved in the production of Humulin are given below. Choose the correct sequence

(i) Synthesis of gene (DNA) for human insulin artificially → (1)

(ii) Culturing recombinant E.coli in bioreactors → (4)

(iii) Purification of humulin → (6)

(iv) Insertion of human insulin gene into plasmid → (2)

(v) Introduction of recombinant plasmid into E.coli → (3)

(vi) Extraction of recombinant gene product from E.coli → (5)

(1) i, iii, v, vi, ii, iv

(2) ii, i, iv, iii, v, vi

(3) iii, v, ii, i, vi, iv

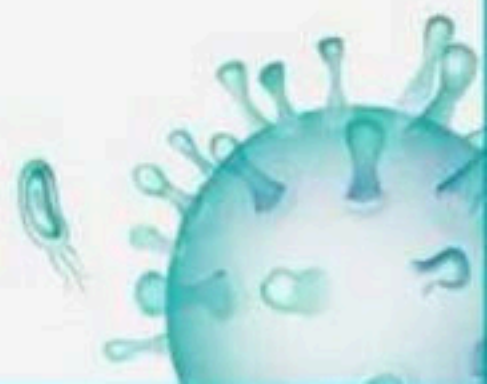
☒ (4) i, iv, v, ii, vi, iii

ii, i, iv, v, ii, vi



7. GMO which are good example of pest resistant crop is

- ✓ (1) Bt-cotton ↩
- (2) Golden rice
- (3) Transgenic tobacco plant with nematode genes ✗
- (4) All



8. Silencing of a specific mRNA translation could be achieved through;

- (1) Antisense RNA ✓
- (2) RNA interference technique ✓
- ✓ (3) Both (1) and (2)
- (4) Microinjection



9. Statement A : Genetically modified crops are ~~less~~ ^{more} expensive than agrochemicals ✗

Statement B : Gene Revolution succeeded in tripping the food supply ✗

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

✓ (4) Both Statements are incorrect.



10. When gene targeting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as:

- (1) Biopiracy
- ✓(2) Gene therapy
- (3) Molecular diagnosis
- (4) Safety testing



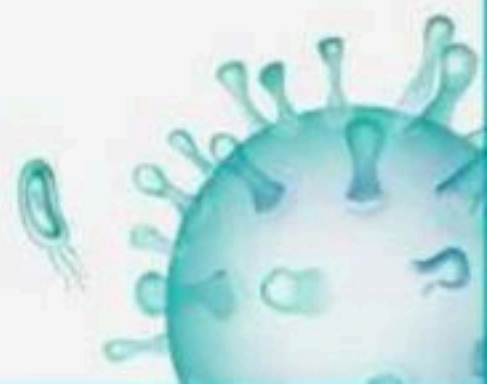
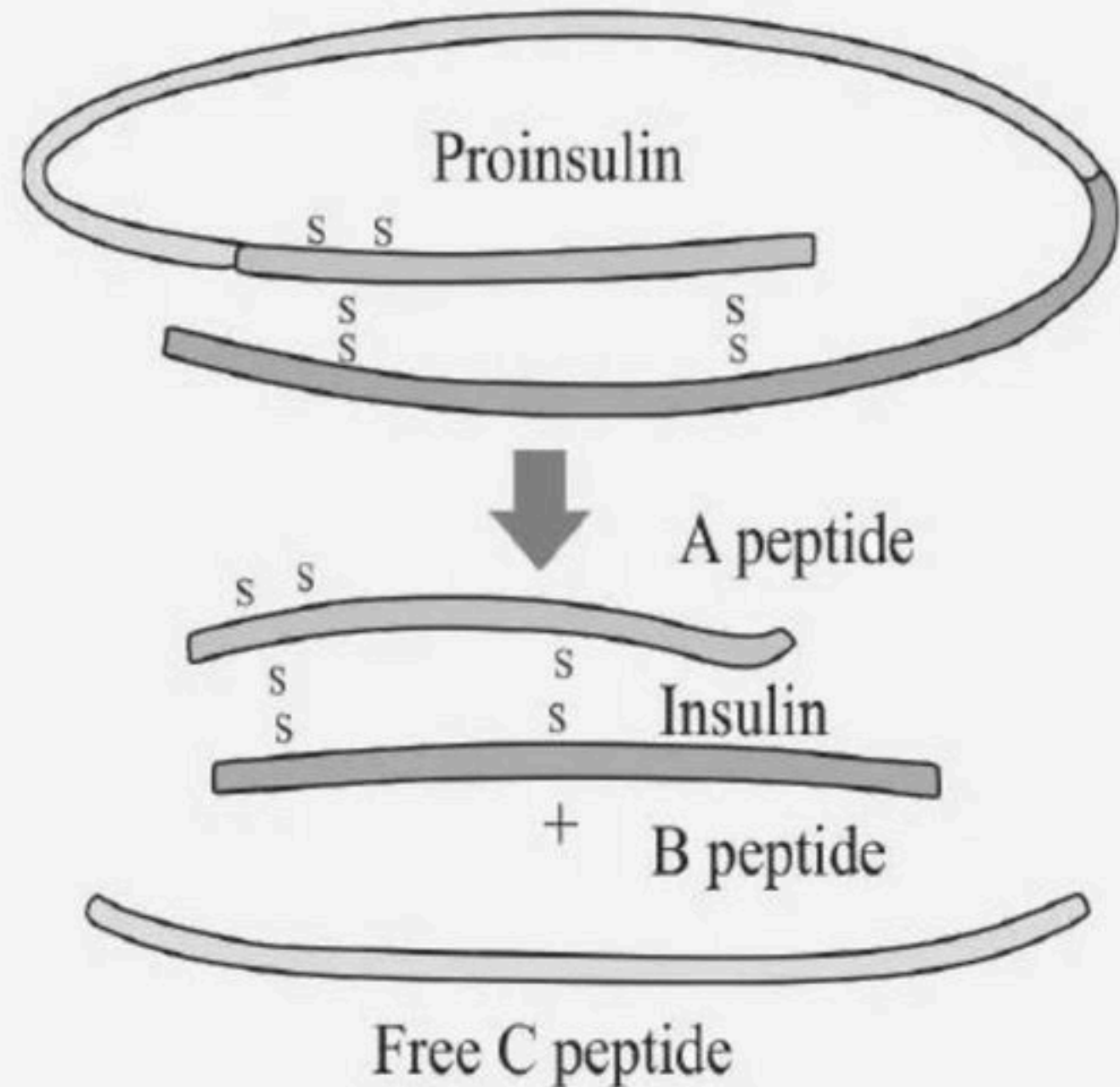
11. What does Bt means?

- ✓ (1) Bacillus thuringiensis
- (2) Barium treated cotton seeds
- (3) Bacteria terminator
- (4) Bacillus tuberculae



12. The diagram depicts

- (1) Breakdown of pro-insulin into insulin
- ✓ (2) Maturation of pro-insulin into insulin
- (3) Maturation of insulin into pro-insulin
- (4) None of the above



13. Assertion : Biotechnology produces transgenic microorganisms that acts as microfactories for proteins.

Reason : To produce proteins of human use like insulin.

Transgenic microorganisms can be developed.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

✓ (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.



14. The site of production of ADA in the body is:-

- (1) Erythrocytes
- ✓ (2) Lymphocytes
- (3) Blood plasma
- (4) Osteocytes



15. Assertion : Humulin is better than conventional insulin

Reason : Conventional insulin produces many side effects

- ☒ (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect.



16. Transgenic animals are produced for which of the following purposes?

- i. To study-how gene are regulated and how they affect
the normal functions of body and its development ✓
- ii. To study of diseases like cancer and phenylketonuria ^x ✗
- iii. To obtain useful ^{biological} chemical product ✗
- iv. To test vaccine safety and chemical safety ✓

(1) All of them

✓ (2) i and iv

(3) ii and iv

(4) Only i



17. RNA interference involves :-

- (1) Interference of RNA in synthesis of DNA
- (2) Synthesis of mRNA from DNA
- (3) Synthesis of cDNA and RNA using reverse transcriptase
- ✓ (4) Silencing of specific mRNA due to complementary
dsRNA



18. Match the following

Column I		Column II	
A	RNAi	P	Right granted for <u>biological</u> entities
B	Biopatent	Q	Gene silencing
C	Gene therapy	R	Effort to <u>fix the non-functional</u> gene
D	Biopiracy	S	Illegal removal of biological materials

- (1) A - R, B - S, C - P, D - Q
☒ (2) A - Q, B - P, C - R, D - S
(3) A - Q, B - S, C - R, D - P
(4) A - R, B - Q, C - P, D - S



19. A novel strategy was adopted to prevent . incognita infection in tobacco plants that was based on the process of

(1) DNA interference

✓(2) RNA interference

(3) RNA replication

(4) DNA replication



20. Which of the following statements is false?

- I. Insulin was originally extracted from pancreas of slaughtered pigs and cattle ✓
- II. Animal insulin is difficult to obtain ✓
- III. Animal insulin is identical to human insulin ✗
- IV. Non human insulin caused some patients to develop allergy ✓
- V. Recombinant DNA allowed scientists to insert a human insulin gene into a bacterial expression sector
- VI. Recombinant insulin is actually obtained from E. coli bacterial cell

(1) Only I and II

(2) Only III and IV

✓ (3) Only III

(4) Only VI



21. Human protein which is being obtained from transgenic animal and used to treat emphysema is

- (1) Trypsinogen
- (2) Alpha lactalbumin
- (3) Trpsin
- ~~(4) α -1-antitrypsin~~



22. Read the following statements and select the correct option

- ✓ I. Recombinant DNA technology is used to improve crop plants by increasing their productivity, by making them more nutritious and by developing disease resistant.
- ✓ II. Bt cotton is resistant to bollworm infestation. *Particular*
- ✗ III. Bacillus thuringiensis form cry protein during any phase of their growth ✗
- IV. Bacillus thuringiensis is not harmed by self Cry protein because of its occurrence as protoxin (inactive) ✓
- V. Protoxin Cry protein is changed into active Cry protein in the stomach of insects due to alkaline in stomach ✓

(1) All are correct

(2) I and IV are correct

✓ (3) Only III is false

(4) All are false



23. Which one is incorrect?

- (1) Gene therapy is a collection of methods that allows correction of a gene defect diagnosed in a child / embryo
- (2) Normal gene is introduced into the individual or embryo to compensate for non-functional gene
- (3) Genes are inserted into a person's cells & tissues to treat a disease
- ✓ (4) The protein encoded by cryII Ab control corn borer.

often hollow



24. Which of the following transgenic animals are used in testing safety of polio vaccine before they are used on human?

- (1) Transgenic cow
- ✓ (2) Transgenic mice
- (3) Transgenic monkey
- (4) Transgenic sheep



25. In genetic engineering, a DNA segment (gene) of interest, is transferred to the host cell through a vector. Consider the following four agents (A-D) in this regard & select the correct option about which one or more of these can be used as a vector / vectors

i. A bacterium

✓ ii. Plasmid

iii. Plasmodium

✓ iv. Bacteriophage

(1) i, ii & iv only

(2) i only

(3) i & iii only

✓ (4) ii & iv only



26. Select the correct answer

i. Using rDNA technique insulin assembled into a mature form, which does not contain any extra stretch called A-peptide. ✗

ii. In 1983, Eli Lilly an American company prepared two DNA sequences corresponding to A and B, chains of human insulin and introduced them in plasmids of E. coli to produce insulin chains. ✓✓

(1) Only i is correct

✓✓ (2) Only ii is correct

(3) Both are correct

(4) None are correct



27. Introduction of transgenes will result in?

- (1) Formation of new species
- (2) Formation of new protein ✓
- (3) Alter the bio synthetic pathway ✓
- (4) ✓ Both 2 & 3



✓ 28. Statement A : ELISA is based on the principle of antigenantibody interaction.

✓ Statement B : The milk containing the human α -lactalbumin is nutritionally a more balanced product for human babies than natural cow milk.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓ (3) Both statements are correct.

(4) Both Statements are incorrect.



29. In RNAi, genes are silenced using:

(1) ss DNA

(2) ds DNA

✓ (3) ds RNA ✓

(4) ss RNA



30. $\alpha - 1$ antitrypsin is

(1) an autacid

(2) an enzyme

✓(3) used to treat emphysema

(4) used to treat arthritis



31. Some of the characteristics of Bt cotton are

- ✓ (1) High yield and resistance to bollworms
- (2) High yield and production of toxic protein crystals
which kill dipteran pests ✗
- (3) Long fibre and resistance to beetle pests
- (4) Long fibre and resistance to aphids



32. Which of the following glucose transporters is insulin independent?

- (1) GLUT I
- (2) GLUT II
- (3) GLUT III
- ✓ (4) GLUT IV



33. Select the incorrect statement :-

(1) Genetic modification of organisms can have unpredictable results when such organisms are introduced into the ecosystem. Therefore the Indian government has set up organization such as GEAC

✓ (2) It ensures the safety of introducing GM organisms for ^{Public}~~private~~ services ✗

(3) It makes decision regarding the validity of GM research

(4) All are correct



34. In RNA interference, the dsRNA molecule prevents :-

- (1) Transcription of mRNA
- (2) Transport of RNA from nucleus to cytoplasm
- ✓✓ (3) Translation of mRNA ✓✓
- (4) Aminoacylation



35. Which of the following statements is / are true about genetically engineered insulin?

I. Its name is humulin ✓

II. It was manufactured by American firm Eli Lilly ✓

III. Animal insulin is not identical to human insulin. ✓

Actually insulin contain A and B peptide chains. ✓

IV. It is produced by the fermentation of appropriate recombinant E. coli clones,

(1) I, II

(2) II, III

(3) I, II, III

✓(4) I, II, III, IV





36 Statements related to human insulin are given below :
Which statement(s) is / are correct about "genetically engineered insulin"

- I. Pro-hormone insulin contain extra stretch of C-peptide ✓ X
- II. A-peptide and B-peptide chains of insulin were produced separately in E. coli,
extracted and combined by creating disulphide bond between them. ✓
- X III. Insulin used for treating Diabetes was extracted from Cattles and Pigs
- ✓ IV. Pro-hormone insulin needs to be processed for converting into a mature and functional hormone.
- X V. Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below:

- ✓ (1) II only (2) III and IV only
- (3) III, IV and V only (4) I, II and IV only



37. Production of giant mouse in the laboratory was possible due to

- (1) Gene synthesis
- (2) Gene duplication
- (3) Gene mutation
- ✓ (4) Gene manipulation



38. What is the collection of methods that allows correction of a gene defect that has been diagnosed in a child/embryo called?

- ✓ (1) Gene therapy ↪
- (2) rDNA technology
- (3) Bone marrow transplantation
- (4) Enzyme replacement therapy



39. Select the correct sequence of steps in the production of transgenic animals.

i. Identification and separation of desired gene → ①

ii. Selection of proper vector → ②

iii. Introduction of transferred vector in cells and tissues → ④

iv. Combining desired gene with vector → ③

v. Demonstration of integration and expression of → ⑤

foreign gene in transgenic tissue

(1) ii, i, iii, iv, v

(2) i, iii, ii, v, iv

(3) ii, i, iv, iii, v

✓(4) i, ii, iv, iii, v



40. Assertion : A crop expressing cry gene is usually resistant to group of insects

Reason : 'cry' proteins produced from Bacillus thuringiensis is toxic to larvae of certain insects

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

☒ (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

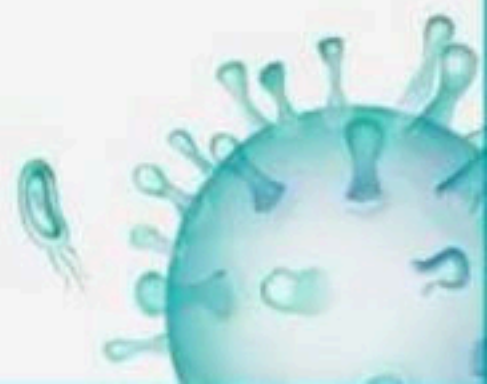


41. Match Column I with Column II:

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Column I		Column II	
A	α -1 antitrypsin	P	Cotton bollworm
B	<i>CryIAb</i>	Q	ADA deficiency
C	<i>CryIAc</i>	R	Emphysema
D	Enzyme replacement therapy	S	Corn borer

- (1) A – Q, B – P, C – S, D – R
(2) A – R, B – P, C – Q, D – S
☒ (3) A – R, B – S, C – P, D – Q
(4) A – Q, B – S, C – P, D – R



42. Which of the following part of the plant is infected by the nematode and reduces the production?

- ✓ (1) Roots of tobacco plant →
- (2) Leaves of tobacco plant
- (3) Roots of cotton plant
- (4) Flowers of cotton plant



43. Viruses with RNA genomes & transposons that replicate via an RNA intermediate are

- (1) Good genetic engineers
- (2) Used as vectors in genetic engineering
- (3) Silenced through RNAi
- ✓ (4) Source of complementary RNA involving in silencing of mRNA in RNA



44. Use of bioresources by multinational companies and organisations without authorization from the concerned country and its people is called

(1) Biodegradation

✓(2) Biopiracy

(3) Bio-infringement

(4) Bioexploitation



45. What is true about insulin?

- (1) It can be administered orally to diabetics
- (2) Insulin from animals may not elicit an immune response in human body
- ✓ (3) In 1983, Eli Lilly an American company prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them in plasmids of E. coli to produce insulin chains.
- (4) Insulin cannot be produced by G.M. bacteria as they are prokaryotic



46. Which is the step taken by Government of India to fulfill the requirement of patent terms and other emergency provisions in this regard?

(1) ETI act

(2) Biopiracy act

✓ (3) Indian Patents bill ←

(4) None of the above



47 Bt strains have been used for designing novel

(1) Biofertilizers

✓ (2) Bioinsecticidal

(3) Bioremediation

(4) Phytoremediation



48. Proteins encoded by gene cryIAb controls

(1) Bollworm

(2) Insects

✓✓ (3) Corn borer ✓✓

(4) Nematodes



49. Genetically modified food is not desirable because

- (1) It may cause allergies ✓
- (2) Could lead to antibiotic resistance in human beings
- (3) Disturbance metabolism ✓
- ✓ (4) All of the above



50. Which one of the following techniques does not serve the purpose of early diagnosis of a disease for its early treatment?

- (1) Recombinant DNA Technology
- ☒ (2) Serum and Urine analysis
- (3) Polymerase Chain Reaction (PCR) technique
- (4) Enzyme Linked Immuno-Sorbent Assay (ELISA) technique



51. Match the columns I and II.

Column I		Column II	
A	GEAC	P	Patent terms emergency provisions
B	Biopiracy	Q	Validity of <u>GM reaserch</u>
C	Indian parents bill	R	Use of <u>bioresources</u> <u>illegally</u>

(1) A – R, B – P, C – Q

☒ (2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



52. Insulin consists of two short polypeptide chains:

chain____ and chain____, that are linked together by____?

(1) A, C, disulphide bridges

(2) A, B, hydrogen bridges

☒ (3) A, B, disulphide bridges

(4) A, C, hydrogen bridges



11.1m

53. *Meloidegyne incognita* belongs to ____.

(1) Trematode- platyhelminthes

(2) Nematode- platyhelminthes

✓ (3) Nematode- aschelminthes

(4) Trematode- aschelminthes



54. The Adenosine deaminase deficiency results into:

- ✓✓ (1) Dysfunction of Immune system ✓✓
- (2) Parkinson's disease
- (3) Digestive disorder
- (4) Addison's disease



55. Biopatent means

- (1) Right to use an invention ✗
- (2) Right to use knowledge about biological resources ✗
- (3) Both 1 and 2
- ✓ (4) None of the above



56. The vector used in RNAi is

- ✓✓ (1) Agrobacterium 
- (2) Bacillus thuringiensis
- (3) Azotobacter
- (4) Yeast



57. RNAi which is employed in making tobacco plant resistant to *Meloidogyne incognita* is essentially involved in

- ✓ (1) Preventing the process of translation of mRNA in nematode
- (2) Preventing the process of transcription
- (3) Preventing the folding of protein into functional state after translation
- (4) Preventing the process of splicing of hnRNA



58. Which of the following statements is incorrect about gene therapy in ADA deficiency?

- ☒ (1) Patient does not require periodic infusion of genetically engineered lymphocytes. **X**
- (2) Lymphocytes are then introduced in the body of patient.
- (3) A functional ADA cDNA using a retroviral vector is introduced into these lymphocytes.
- (4) Lymphocytes from patients' blood are taken out and cultured.



59. Which of the following statements is/are correct about Adenosine Deaminase (ADA) deficiency?

I. In the absence of adenosine deaminase enzyme, ATP metabolism is disturbed and neutrophils fail to function. ✗

II. ADA deficiency is caused by the addition of the gene for ADA. ✗

III. ADA patients can be permanently cured by bone marrow transplantation and enzyme replacement therapy. ✗

IV. For permanent cure, the faulty genes are replaced with the genes isolated from the bone marrow cells producing ADA is introduced into cells in any stage ✗

(1) I, II and III

(2) II and IV

(3) I, III and IV

✓ (4) None of these



60. In gene therapy of Adenosine Deaminase (ADA) deficiency, the patient requires periodic infusion of genetically engineered lymphocytes because:

- (1) Gene isolated from marrow cells producing ADA is introduced into cells at embryonic stages.
- (2) Lymphocytes from patient's blood are grown in culture, outside the body.
- ✓(3) Genetically engineered lymphocytes are not immortal cells. ✓
- (4) Retroviral vector is introduced into these lymphocytes.



61. Select the correct answer

i. Silencing of an unwanted gene - DNA polymerase ✗

ii. RNAi - Cellular defence ✓✓

(1) Only i is correct

✓✓ (2) Only ii is correct

(3) Both are correct

(4) None are correct



62. The two polypeptides of human insulin are linked together by

- (1) Phosphodiester bond
- (2) Covalent bond
- ✓✓ (3) Disulphide bridges ✓✓
- (4) Hydrogen bridges



63. Statement A : RNAi is a cellular defence mechanism in prokaryotes.

Statement B : It is silencing of a specific ~~X~~^m-RNA.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- ✓ (4) Both Statements are incorrect.



**64. Statement A : Recombinant therapeutics do not induce unwanted Immunol-
ogical responses.**

**Statement B : Animals whose DNA is manipulated to
possess and express an extra (foreign) gene are known as Transgenic animals**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓(3) Both statements are correct. ✓

(4) Both Statements are incorrect.



65. Statement A : ADA deficiency disorder is caused due to the excessive ^X synthesis of gene for adenosine deaminase.

Statement B : It affects the human immune system. ✓

- (1) Statement A is correct but Statement B is incorrect.
- ✓ (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



66. Biopiracy is related to?

- (1) Traditional knowledge exploitation ✓
- (2) Biomolecules and bioresources exploitation ✓
- (3) Stealing of traditional knowledge about bio resource
- (4) ✓ All of the above



67. Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called

(1) Recombinant organisms

(2) Hybrid organism

✓ (3) Genetically modified organisms

(4) Super-organisms



68. Which substance is tested in case of toxicity/ safety testing using transgenic animals? → chemicals

✓ (1) Chemicals ↩

(2) Pathogen

(3) The amount of DNA in the cell

(4) The amount of tolerable radiation levels of an organism



69. Bt toxin kills insect by

- (1) Causing nervous problems
- (2) Generating excessive heat
- (3) Inhibiting protein synthesis
- ✓ (4) Creating pores in the midgut epithelial cell to cell swelling and lysis



70. Assertion : Transgenic plant is a product of plant tissue culture.

Reason : An organism that contains and expresses a transgenic organism.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

✓(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.



71. Now a days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hybridise its complimentary DNA in a clone of cells, followed by its detection using autoradiography because:

- (1) Mutated gene partially appears on a photographic film.
- (2) Mutated gene completely and clearly appears on a photographic film.
- ✓ (3) Mutated gene does not appears on a photographic film as the probe has no complementarity with it.
- (4) Mutated gene does not appear on photographic film as the probe has complementarity with it.



72. Among the following which characteristics is applicable to Bt-cotton?

- (1) Bt is abbreviated term for bacterial toxin
- ✓ (2) Such a cotton is resistant to army worm and beetles
- (3) The toxin is activated in the plant body just before it enters in insect.
- (4) The toxin is coded by a gene called 'cry' & inserted in plants by E.coli vector



73. Statement A : Transgenic animals are those which have foreign ^{DNA} ~~RNA~~ in all of their cells ✗

Statement B : Of all the existing transgenic animals, are rats

mice ^{asy.}

- (1) Statement A is correct but Statement B is incorrect.
- ✓ (2) Statement A is incorrect but Statement B is correct. ✓
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



74. Bt toxin is harmful to insects like?

(1) Lepidopterans

(2) Diptera *vs*

(3) Coleopterans

☒ (4) All



75. Identify the correct statement w.r.t. development of transgenic animals for study of diseases

- (1) Transgenic animals are designed to increase our understanding of role of genes in this disease ✓
- (2) Animals are developed as models for human diseases to investigate new methods of treatment
- (3) Models exist for Alzheimer's and cystic fibrosis ✓
- ✓ (4) All of these



76. Which bacterium was the first to be used as biopesticides?

- ✓✓ (1) *Bacillus thuringiensis*
- (2) *Escherichia coli*
- (3) *Nitrosomonas*
- (4) *Agrobacterium tumefaciens*



77. Statement A : Bacillus tumefaciens is popularly used in genetic engineering because it can deliver genes to dicots ✗

Statement B : Using Agrobacterium as vector, the genes can be transferred in biotechnological techniques. ✓

(1) Statement A is correct but Statement B is incorrect.

✓(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.



78. Match the columns I and II.

Column I		Column II	
A	ELISA	P	Amplification of nucleic acid
B	PCR	Q	A single stranded DNA or RNA, tagged with a radioactive molecule
C	Probe	R	Antigen-antibody reaction

- ✓ (1) A – R, B – P, C – Q,
(2) A – Q, B – R, C – P,
(3) A – P, B – R, C – Q,
(4) A – Q, B – P, C – R



79. Which Indian plants have either been patented or attempts have been made to patent them by Western nations for their use?

- (1) Basmati rice ✓
- (2) Neem ✓
- (3) Turmeric ✓
- ✓ (4) All of these



80. A doctor while operating on an HIV + patient accidentally cut himself with a scalpel. He comes to you, suspecting himself to have contracted the virus. Which test will you advise him to rule out/confirm his suspicion?

- ☒ (1) PCR
- (2) Routine urine examination
- (3) TLC
- (4) DLC



81. Transgenic animals are those which are

(1) Foreign DNA in some of its cells

✓ (2) Foreign DNA in all its cells ✓✓

(3) Foreign RNA in all its cells

(4) both (1) and (3)



82. Assertion : 'Cry' proteins are named so because they are crystal proteins. ✓

Reason : In alkaline environment of insect midgut 'Cry' proteins are solubilized and then release toxic core fragments after proteolytic action. ✓

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

✓ (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.



83. Tobacco plant resistant to a nematode have been developed by the introduction of DNA, which is produced (in the host cells) as

- (1) Only sense RNA
- ✓ (2) Both sense and antisense RNAs ✓✓
- (3) Only antisenses RNA
- (4) Toxic protein



84. Which of the following are the critical research areas of biotechnology?

- (1) Providing best catalyst ✓
- (2) Creating optimal conditions for catalyst function ✓
- (3) Downstreaming processing technique ✓
- ✓ (4) All of the above



85. Statement A : The inactive protoxin gets converted into active form in the insect gut

Inactive.

Statement B : Bt proteins exist as ~~active~~ toxins in the Bacillus

- ☒ (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



86. Example for gene therapy is :-

- (1) Production of test tube babies by artificial insemination and implantation of fertilized eggs
- (2) Production of injectible hepatitis B vaccine
- ✓ (3) Introduction of gene for adenosine deaminase in persons suffering from SCID
- (4) Production of vaccines in food crops like potatoes which can be eaten



87. Statement A : Genetically engineered crop-based agriculture is used for increasing food production

Statement B : Genomes of many plants are manipulated or altered by combining them with other genes to get desired traits.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓ (3) Both statements are correct. ✓

(4) Both Statements are incorrect.



88. Statement A : A crop expressing cry gene is usually resistant to specific insects ✓

Statement B : cry' proteins produced from Bacillus thuringiensis is toxic to certain insects ✓

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓ (3) Both statements are correct.

(4) Both Statements are incorrect.



89. Use of genetically modified crops in crop field may

- (1) Crops more tolerant to abiotic stresses ✓
- (2) Maximise yield ✓
- (3) Reduce post harvest losses. ✓
- (4) All of the above ✓



90. Technique used to detect mutated genes is called :-

- (1) Gel electrophoresis
- (2) Polymerase chain reaction
- (3) Gene therapy
- ☒ (4) Autoradiography ☒

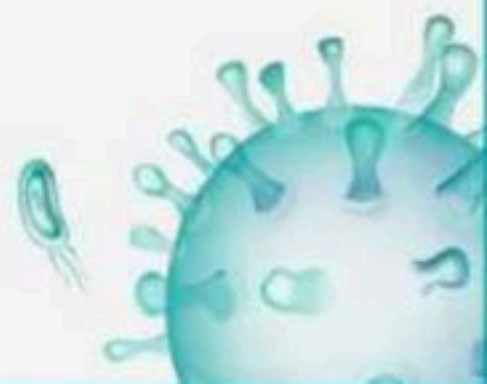


91. With regard to insulin choose correct options. ∴-

- 1) C-peptide is not present in mature insulin ✓
- 2) The insulin produced by Rdna technology has Cpeptide. ✗
- 3) The pro-insulin has C-peptide ✓
- 4) A-peptide and B-Peptide of insulin are interconnected by disulphide bridges. ✓

Choose the correct answer from the options given below.

- | | |
|-----------------------------|----------------------|
| (1) (2) and (4) only | (2) (2) and (3) only |
| ✓ (3) (1), (3) and (4) only | (4) (1) and (4) only |



92. Choose the correct statements for molecular diagnosis of diseases.

I. Using conventional methods of diagnosis (serum and urine analysis, etc), early detection is not possible. ✓

II. By the time when the pathogen has produced a disease symptom, the concentration of pathogen is not really high in the body. ✗

✓ III. Molecular probe is a single-stranded DNA or RNA tagged with a radioactive molecule.

IV. ELISA is based on the principle of antigen-~~antigen~~ interaction. ✗

(1) II and IV

✓ (2) I and III

(3) I and II

(4) II and III



93. Statement A : The two polypeptide chains are interconnected by disulphide bridges ✓

Statement B : In mammals including humans, insulin is synthesised as a prohormone, which contains an extra stretch called the Cpeptide ✓

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓(3) Both statements are correct.

(4) Both Statements are incorrect.



94. Choose the correct statements about PCR technique

I. It is used to detect HIV in suspected AIDS patients. ✓

II. It is used to detect mutations in the genes in suspected cancer patients. ✓

III. Molecular probe shows complementary with mutated gene. ✗

IV. It is a good technique to identify many other genetic disorders. ✓

(1) I and II

✓(2) I, II and IV

(3) III and IV

(4) II, III and IV



95. Match the following

Column I		Column II	
A	<i>Meloidogyne incognita</i>	P	Controls corn borer
B	<i>cryIAb</i>	Q	Controls bollworm
C	<i>cryIAc</i> and <i>cryIIAb</i>	Q	Controls bollworm
D	ADA deficiency	R	Gene therapy
		S	Infects roots of tobacco plant

- (1) A - S, B - P, C - Q, D - R
(2) A - P, B - R, C - S, D - Q
(3) A - Q, B - P, C - R, D - S
(4) A - S, B - Q, C - R, D - P



96. Assertion : Cellular defence mechanism in eukaryotes is RNAi.

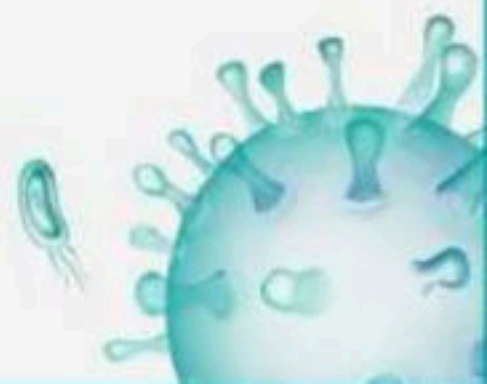
Reason : RNAi is silencing of a specific ~~X~~^mRNA.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

✓(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.



97. Statement A : 'Silencing of mRNA molecule' in order to control the production of a harmful protein has been used in the protection of plants from armyworm ✗

Statement B : ✗ 1-antitrypsin gene was introduced in the first transgenic cow ✗

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

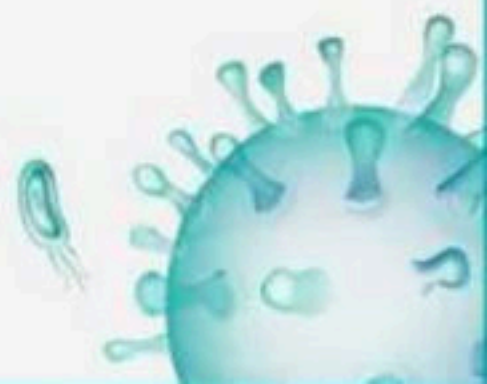
(3) Both statements are correct.

✓ (4) Both Statements are incorrect.



98. Organic agriculture is a technique of raising crops for

- (1) Increased food production
- ✓ (2) Reduced use of chemical based fertilisers and pesticides
- (3) Increasing the use of agrochemicals
- (4) None of the above



99. Assertion : Tobacco plant yield is highly reduced because of damage to roots.

Reason : Nematode Meloidogyne incognita infects the root.

✓✓ (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.



100 GEAC stands for

- (1) Genetic Engineering Authorative Committee
- ✓ (2) Genetic Engineering Approval Committee ✓
- (3) Genetic Engineering Appraisal Committee
- (4) Genetic Engineering Approval Council



THANK YOU



BIOLOGYLIVE



Thank You

8pm